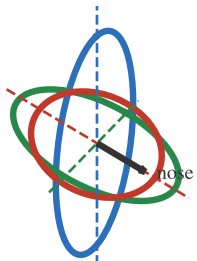
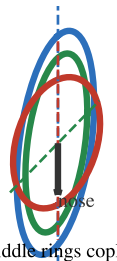
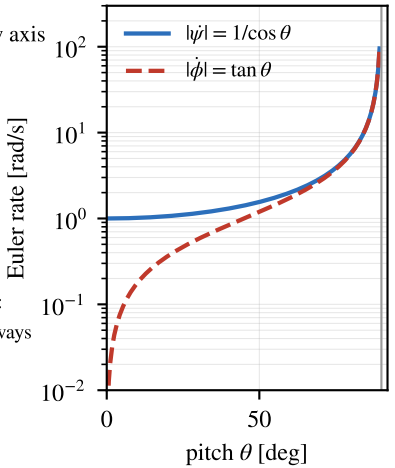
(a) $\theta = 25^\circ$: three independent axes



blue: yaw green: pitch red: roll

(b) $\theta = 90^\circ$: roll axis $\parallel$ yaw axis



outer & middle rings coplanar:
no ring can turn the nose sideways

(c) Euler rates for $\omega_b = (0, 0, 1)$ rad/s